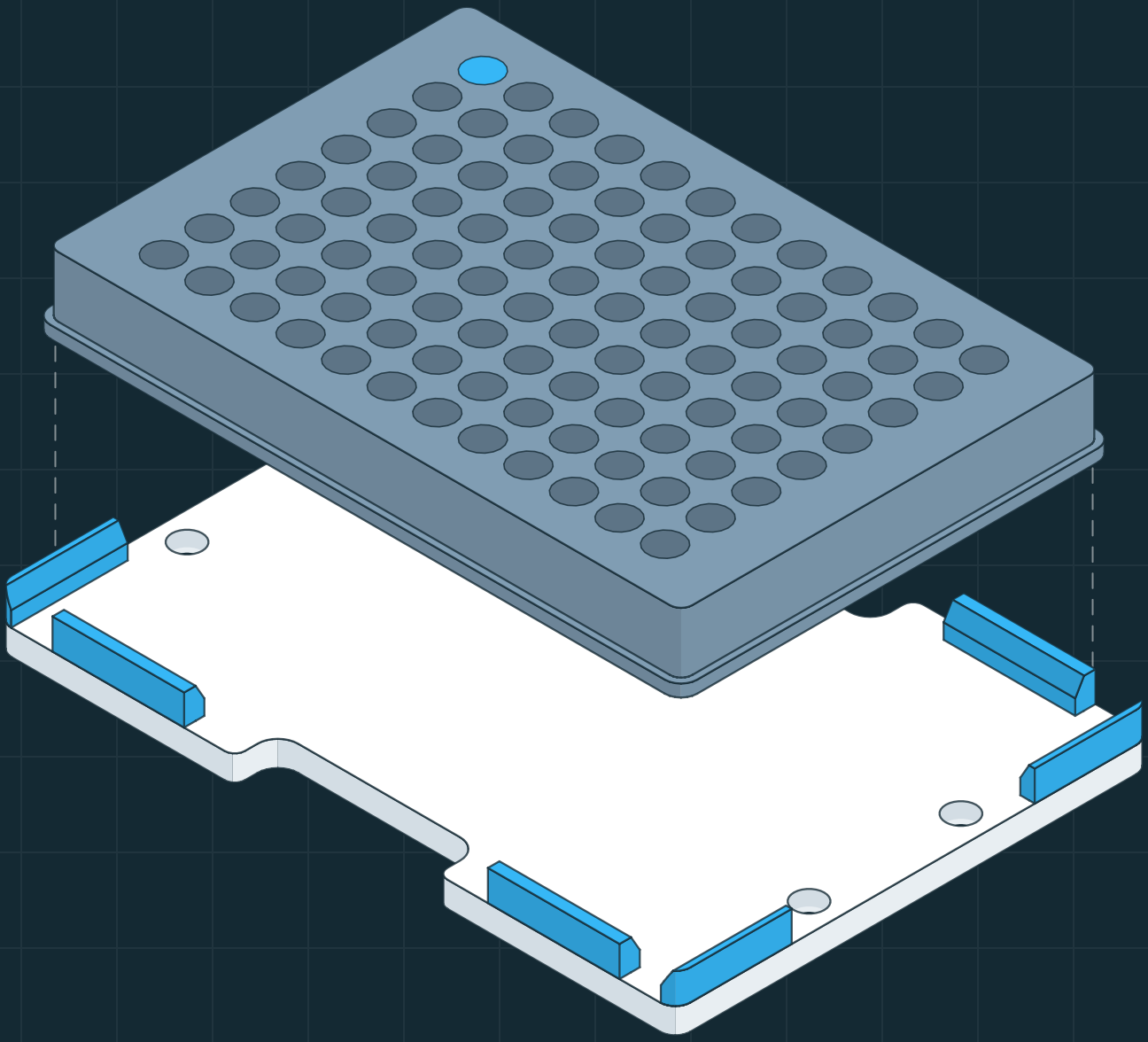
OpenNest

OPEN HARDWARE

Design and manufacturing guide

V1.4 NOVEMBER 2026



OPEN HARDWARE / AGENT READY / ANSI/SLAS 1-2004

Adaptyv

OPENNEST

A nest for every SBS plate.

OpenNest is an open hardware deck plate that holds a standard microplate where a robot can find it. Tell an agent your hole pattern, then print or machine the file it builds.

134 × 94	128.3 × 86	2.0	30°
mm plate, 11 mm tall	mm pocket	mm catch per side	lead-in from vertical

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MODEL FILES

STEP for machining. STL and 3MF for printing.

BUILD SCRIPT

An agent runs it to place your holes.

THIS GUIDE

Drawing, fit limits, process settings and checks.

STATUS

A v1.3 plate has been printed. This v1.4 geometry has not yet been printed, machined or run on a robot. Numbers in this guide are calculated from the model and the SLAS standard. Send us your fit report.

01

Every surface has a job.

A robot places a plate with some error. The nest absorbs that error, sets the plate down flat and leaves room for the gripper to get out.

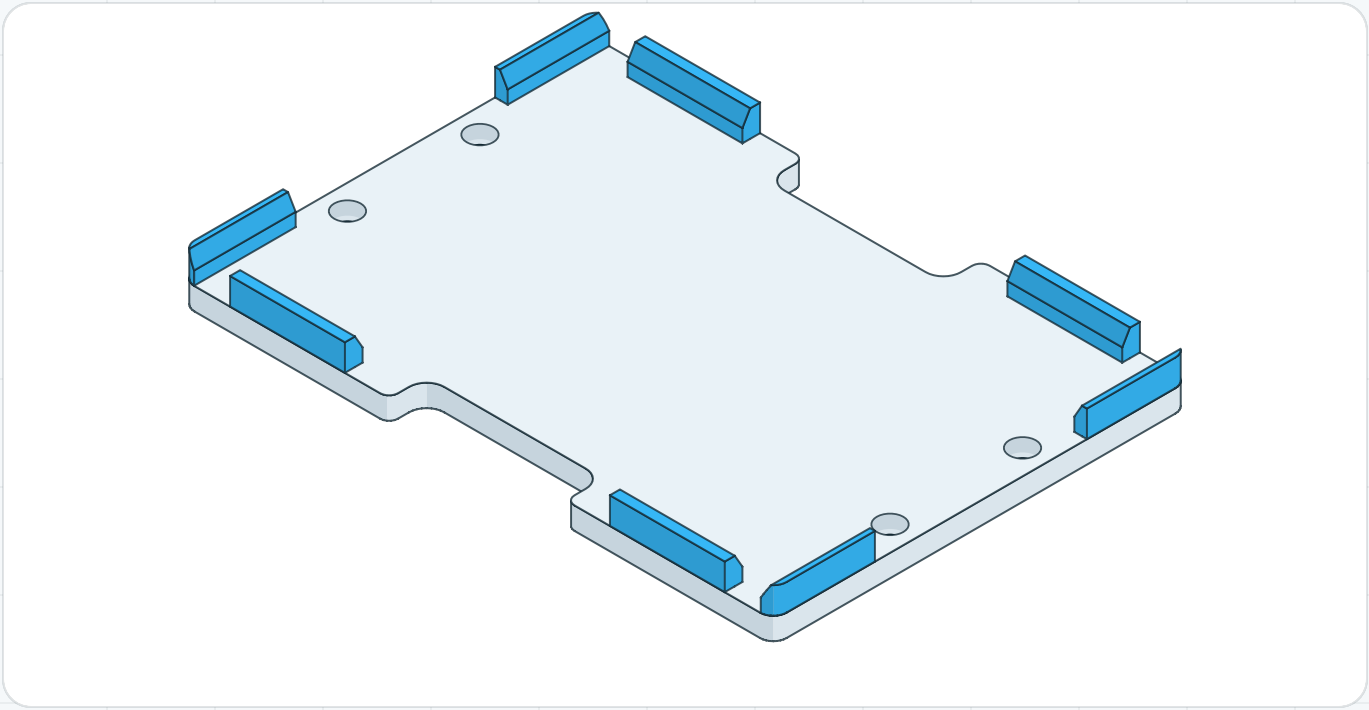


FIG 1 / THE PLATE, DRAWN TO SCALE FROM THE MODEL

Catch

Eight tabs lead a plate in with a 30° slope. The slope starts 3 mm above the seat, so a nominal plate landing up to 2.0 mm off centre still finds the pocket.

Seat

The walls meet the seat in a square corner. Nothing lifts the edge of the plate, so its height stays the height you taught.

Release

Two 40 × 8.5 mm notches on the long sides give gripper fingers somewhere to go. The tabs stop short of them.

Mount

Holes are yours to define. Screw heads sit below the seat, clear of the plate.

BEFORE YOU RUN

Teach each nest on your own deck. Screw clearance lets a plate sit slightly differently from one position to the next, so do not copy positions between nests.

02

Drawing and tolerances.

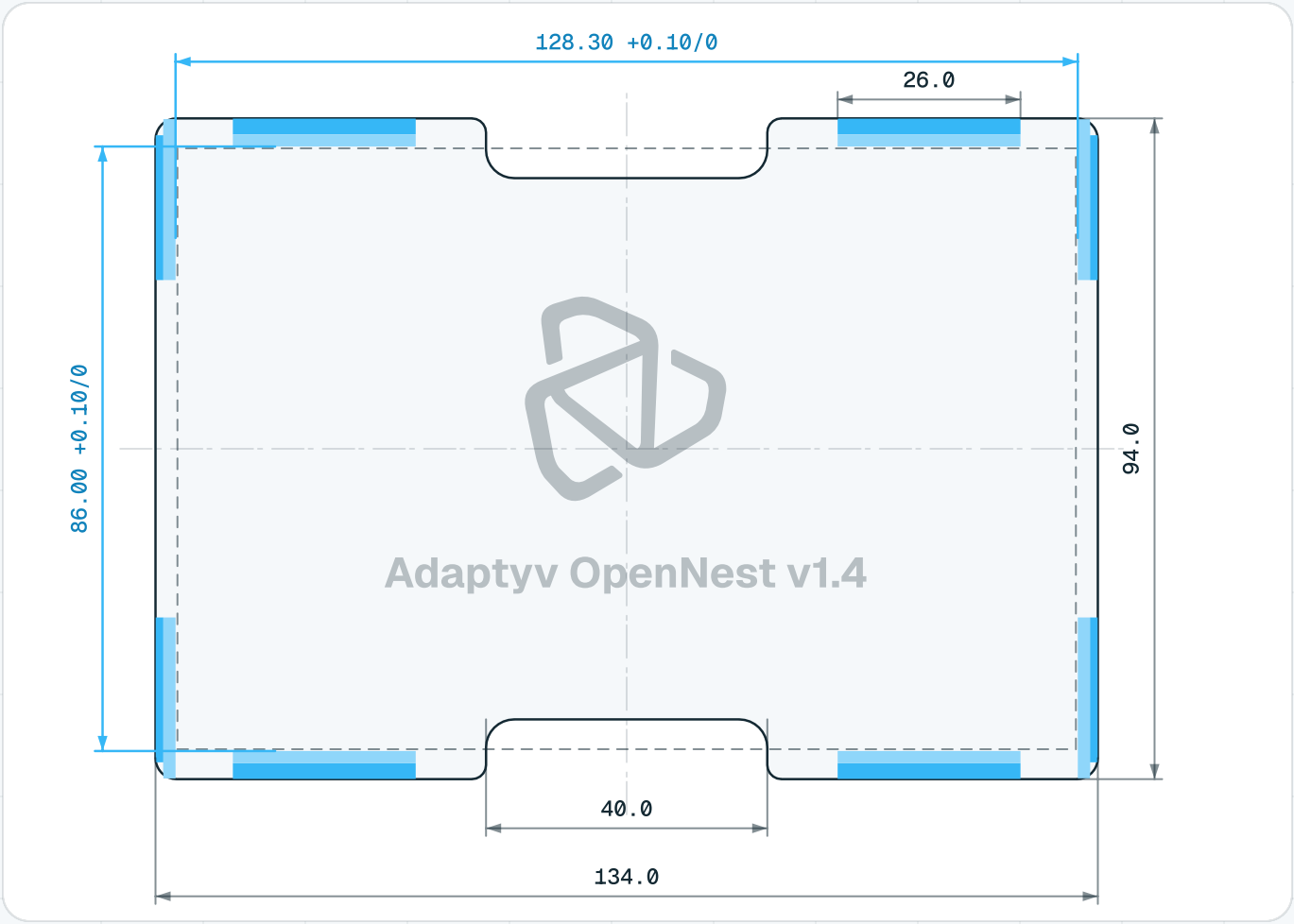


FIG 2 / PLAN VIEW. CYAN DIMENSIONS CONTROL THE FIT. DASHED LINE IS THE SLAS FOOTPRINT

FEATURE	NOMINAL	TOLERANCE
Pocket, X	128.30	+0.10/0
Pocket, Y	86.00	+0.10/0
Seat height from underside	5.00	±0.05
Seat flatness		0.05
Everything else		±0.1, ISO 2768-f

HOLES

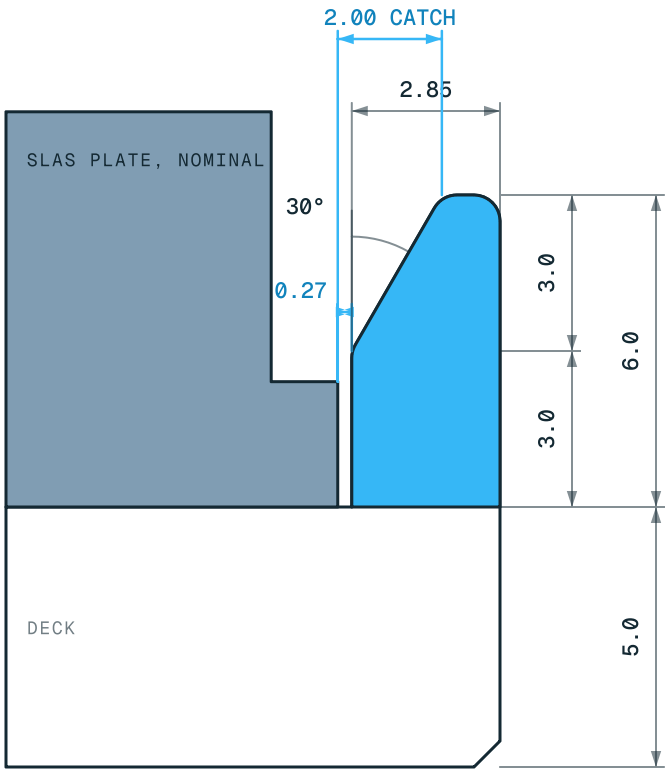
Hole positions are yours to define, so they are not on this drawing. The STEP file is the master.

03

Fit against SLAS 1-2004.

Every plate the standard allows fits at 20 °C. Clearance is total, in mm.

PLATE WITHIN SLAS	SIZE, X × Y	X CLEARANCE	Y CLEARANCE
Nominal	127.76 × 85.48	+0.54	+0.52
Largest within 12.7 mm of a corner, ±0.25	128.01 × 85.73	+0.29	+0.27
Largest anywhere along a side, ±0.5	128.26 × 85.98	+0.04	+0.02



2.00 mm
Catch per side in X, 1.99 mm in Y

±0.27 mm
Plate float in X, ±0.26 mm in Y

30°
Lead-in from vertical

FIG 3 / SECTION THROUGH A CORNER TAB, NOMINAL PLATE AT 20 °C

1536-WELL PLATES

Float of ±0.27 mm is comfortable for 96 and 384 wells. Check your tip and well tolerance before you run 1536-well plates in this nest.

04

3D printing.

A print proves the geometry, not the material. Measure it before you trust it.

SETTING	VALUE
Process	0.12 mm Fine, Bambu H2D, 0.4 mm nozzle
Lead-in, 7.88 to 11 mm above the bed	0.08 mm layers
Walls and infill	4 walls, 25% gyroid
Precise outer wall	On
Seam	Blocked on the tab faces a plate touches
Brim	Outer, 5 mm. Remove before measuring
Ironing	Off
Slicer estimate	About 2 h 45 min, about 50 g PLA

Check the print

01 Let it cool and remove the brim.

02 Measure the pocket between opposing tab walls, at the seat and just under the slope. Accept X 128.1 to 128.5 and Y 85.8 to 86.2.

03 If it is small, scale the object in XY by target / measured in the slicer and print again. A plate that only fails in an undersized printed pocket says nothing about a machined one.

04 Measure the seat height from the underside. The target is 5.0 mm. Check the seat is flat.

05 Then test what a print can prove: plates drop in along the lead-in and sit flat, holes line up with your deck, and the gripper clears.

PLA

PLA shrinks as it cools and softens near 60 °C. Use it for fit checks and room temperature work.

05

CNC machining.

The STEP file is the part. A machine cuts it as drawn, with no shrink correction.

Aluminium

STEP	WHAT TO DO
Stock	12 mm rolled plate, EN AW-6082-T651 or 6061-T651. Not 5083 cast plate, which anodises grey and blotchy.
Op 1, top	Clear the seat around the tabs. Finish the tab walls with a square end mill. Cut the 30° lead-in, drill your holes, then profile the outline and notches.
Op 2, flip	Break the underside edge 0.5 mm and remove the holding tabs.
Finish	Anodize per MIL-PRF-8625, Type II, Class 2. Dimensions apply after coating.
Engraving	0.3 mm deep, engraved before or after coating. Say which on the order.

HARDCOAT

R0.5 edges suit Type II. Thick Type III needs larger edge radii, so change the model before you order it.

Other materials

MATERIAL	POCKET	WHAT TO KNOW
POM-C, PEEK	Size for expansion	Polymers creep under screw heads. Keep torque low and recheck it.
PTFE-impregnated anodise on aluminium	As aluminium	Low friction surface that keeps aluminium's stability. Check the edge radii first.
Solid PTFE	Own geometry	Changes about 0.4 to 0.6 mm over the pocket between 19 and 23 °C, more than the clearance. Needs a larger pocket, a thicker floor and metal sleeves.

06

Agent ready.

The pocket, tabs and notches stay fixed. An LLM agent reads the spec, runs the script with your hole pattern and writes your files.

EXAMPLE PROMPT

```
Make me an OpenNest with four M3 cap head screws,
110 × 60 mm apart. I'm printing it in PLA on a Bambu H2D.
```

Tell the agent

ITEM	WHAT TO GIVE IT
Holes	Positions from the plate centre, or the spacing between them.
Screws	Size and head: cap head or flat head.
Process	3D print, CNC aluminium or another CNC material.
Material	What you will make it from, so it can size the pocket.

How it runs

01 Open the repo in your coding agent.

02 Describe your holes and your process.

03 The agent runs the script, then checks that the holes clear the tabs and the plate.

04 Measure your deck before you cut. An agent can only check holes against the plate, not against your hardware.

FIXED BY DESIGN

Ask for a change to the pocket, lead-in or seat height and you leave the reference geometry. Say so in your fit report.

07

Acceptance check.

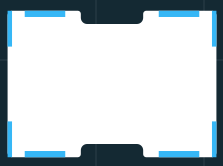
Fill this in for each part before it goes on a deck.

PART	MATERIAL	MEASURED BY	DATE

CHECK	TARGET	MEASURED
Pocket X, at the seat	128.30 +0.10/0	
Pocket Y, at the seat	86.00 +0.10/0	
Pocket X, printed part	128.1 to 128.5	
Pocket Y, printed part	85.8 to 86.2	
Seat height from underside	5.00 ±0.05	
Seat flatness	0.05	
Envelope	134 × 94 × 11	
Plate drops in by hand without sticking	Pass	
Plate sits flat and does not rock	Pass	
Screw heads sit below the seat	Pass	
Holes line up with your deck	Pass	
Robot places and picks without contact	Pass	

SHARE IT

Open an issue with your numbers, printer or shop, and material. Real measurements are what this design needs next.



Build it. Measure it. Share what you find.

GITHUB.COM/ADAPTYVBIO/OPEN-NEST

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